

PAPER_968: Production Scaling v11 Benchmark (500k calc/s)

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Abstract

We benchmark the UQFF production pipeline at 500,000 calculations per second, an 11.1% improvement over v10 (450k). Two new kernels — 26D Ramanujan summation and Triadic solver — bring the total to 16 simultaneously benchmarked kernels.

1. Scaling History

Version	Target (calc/s)	Kernels	Session
v4	100,000	4	198
v5	150,000	6	200
v6	200,000	8	204
v7	300,000	10	208
v8	350,000	11	210
v9	400,000	12	213
v10	450,000	14	214
v11	500,000	16	215

2. New v11 Kernels

kernel_ramanujan_26d_sum

$S_{26}(z) = \Sigma z^n / n^{26} \cdot R_n^{(26)}$ — accelerated polylog for hot-loop evaluation.

kernel_triadic_solver

Combined Compressed + Resonant + Buoyancy Triadic evaluation in a single kernel call.

3. Throughput

$$\text{rate} = \frac{N_{\text{iter}} \times 16}{t_{\text{elapsed}}} \geq 500,000 \text{ calc/s}$$

References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)